

RII Childhood Inequities in Sedentarism

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Aim:

Examine trends in socioeconomic inequalities in leisure-time sedentarism among children aged 6 to 12 years in Spain between 2003 and 2023.

Methods:

Data source: Five cycles (2003, 2006, 2011, 2017, 2019) of the Spanish National Health Survey. Final study population of 19,500.

Exposure: The continuous socioeconomic rank variable derived from occupational social class of the household reference person.

Outcome: Children reported as engaging in any physical activity regardless of frequency were classified as non-sedentary, and those reported as performing none as sedentary.

Statistical Analysis: Quantification of socioeconomic inequalities in childhood sedentarism:

- Poisson regression models with log link and robust standard errors, adjusting for age and sex, incorporating survey weights, representing the Relative Index of Inequality - RII, interpreted as the prevalence ratio of sedentarism between the lowest and highest socioeconomic positions.
- Poisson regression models with identity link, adjusting for age and sex, incorporating survey weights, representing the Slope Index of Inequality - SII, interpreted as the absolute difference in predicted sedentarism prevalence, in percentage points, between the lowest and highest socioeconomic positions.
- Multi-level Poisson regression models with a log link were fitted including all survey years, adjusting for age and sex and incorporating survey weights. Individuals were nested within survey years and NUTS regions. Random intercepts and random slopes for social class were specified at the survey and region levels, allowing baseline sedentarism and socioeconomic inequalities to vary across time and regions.
- Regional clustering was assessed by estimating intraclass correlation coefficients (ICCs) from multilevel Poisson regression models with a log link including random intercepts and slopes for social class.
- The annual percent change in sedentarism inequality overall and separately for each NUTS region from 2003 to 2023 was calculated, where positive APC indicate increasing RII over time.
- The inflection point in the temporal trend of sedentarism inequality between 2003 and 2023 was identified using segmented regression.

NUTS regions classification:

- The Canary Islands; Centre: Castile and Leon, Castilla-La Mancha, and Extremadura; East: Catalonia, the Valencian Community, and the Balearic Islands; Madrid: the Community of Madrid; North-East: the Basque Country, Navarre, La Rioja, and Aragon; North-West: Galicia, Asturias, and Cantabria; and South: Andalusia, Murcia, and the autonomous cities of Ceuta and Melilla.

Results:

Descriptive sample characteristics: The sex, municipality type, and region distribution remained stable across survey years. Boys represented approximately half of the sample each year. Municipality type remained relatively consistent across survey years, with approximately half of the children living in urban municipalities, followed by semi-urban and rural areas. The regional distribution was also stable, with the East and South regions representing the largest proportions across all years. The age, nationality and occupational social class distributions varied across survey years. Children aged 6 to 9 years represented the largest proportion in most years. The proportion of children with foreign nationality increased steadily over time. Occupational social class distributions shifted over time, with an increase in the proportion of children from lower occupational social classes, particularly Class V.

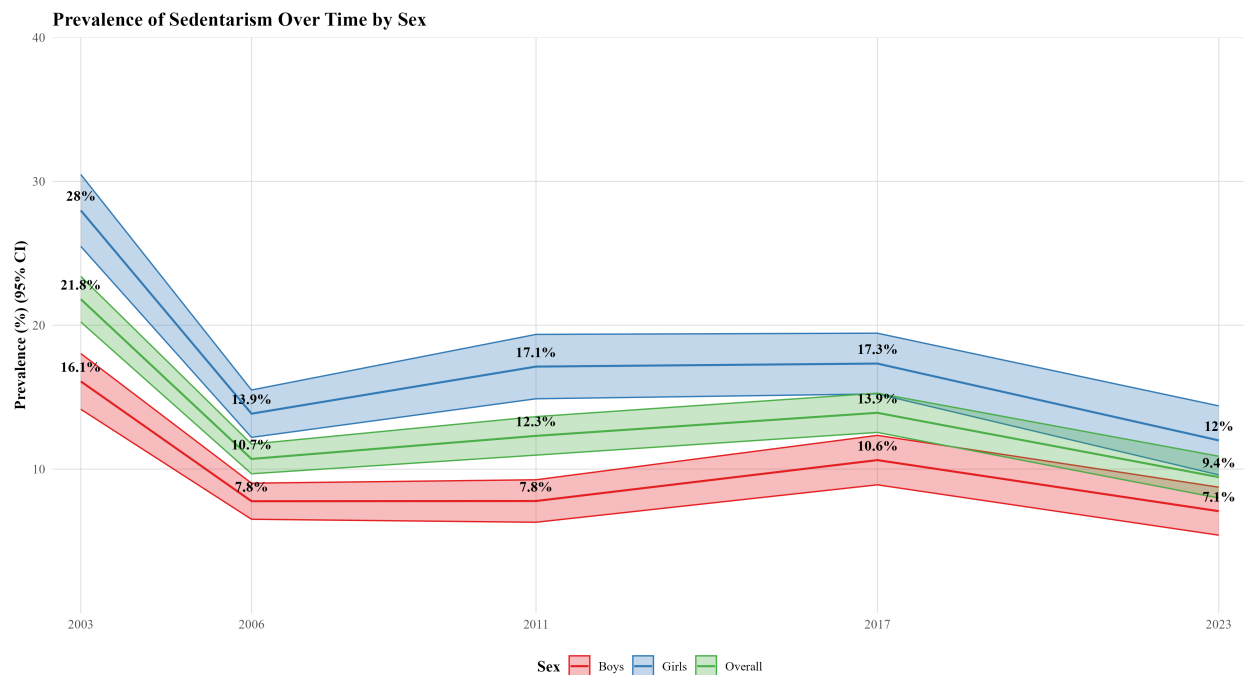


Figure 1: Prevalence of Sedentarism 2003-2023 by Sex

Sedentarism prevalence across survey years: Overall decrease, 21.8% (95% CI: 20.2, 23.4) in 2003 to 9.4% (95% CI: 8.0, 10.9) in 2023. Among girls, 12.0% (95% CI: 9.6, 14.4) and boys, 7.1% (95% CI: 5.4, 8.7) in 2023. Among the lowest socioeconomic position, 20.4% (95% CI: 14.8%, 26.0%), and among the highest, 5.8% (95% CI: 2.6, 9.0) in 2023.

Inequalities in sedentarism across survey years: RII rose from 1.82 (95% CI: 1.50, 2.22) in 2003 to 5.59 (95% CI: 3.42, 9.08) in 2023, indicating that children in the lowest social class had more than five times the prevalence of sedentarism compared to those in the highest. In 2023, inequalities were slightly greater among girls at 5.90 (95% CI: 2.47, 11.50) than boys at 5.35 (95% CI: 3.17, 10.99).

SII, the predicted prevalence of sedentarism, was 13.01 (95% CI: 8.97, 17.05) percentage points higher among children in the lowest social class compared to the highest in 2003 and remained similar in 2023 at 12.37 (95% CI: 8.42, 16.33). Among girls, this absolute inequality reached 16.83 (95% CI: 10.45, 23.21) percentage points in 2023 and peaked at 20.73 (95% CI: 15.48, 25.98) percentage points in 2017.

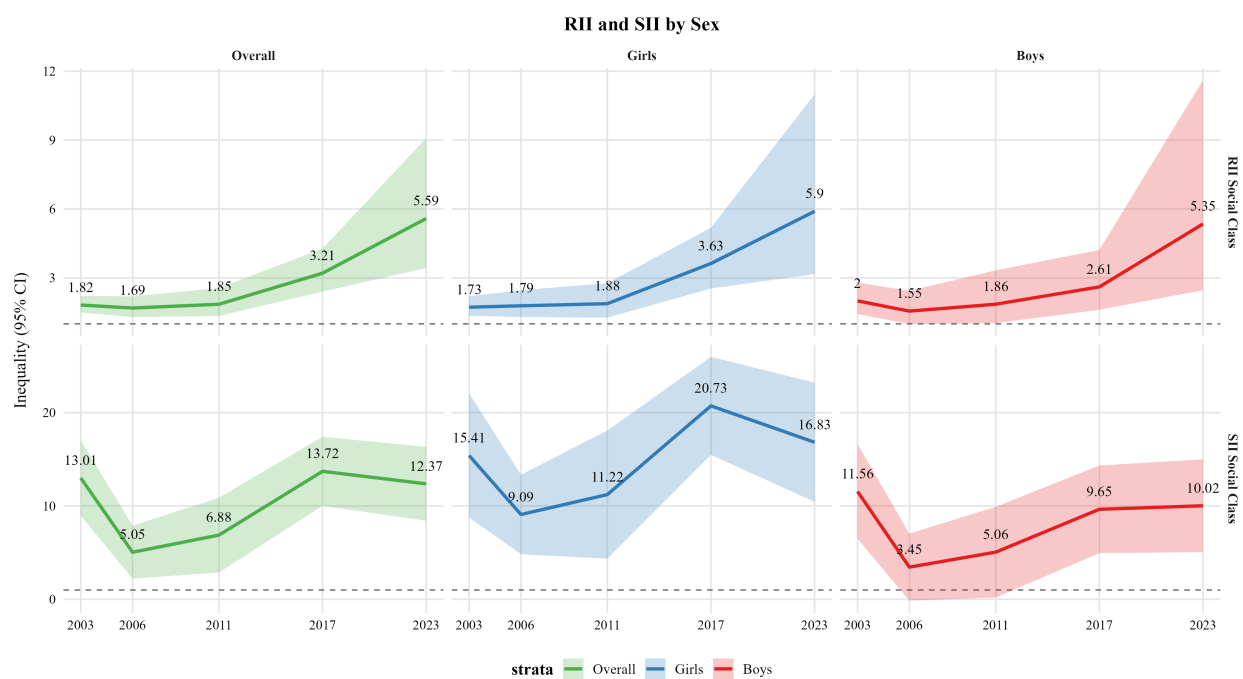


Figure 2: RII and SII overall and by sex

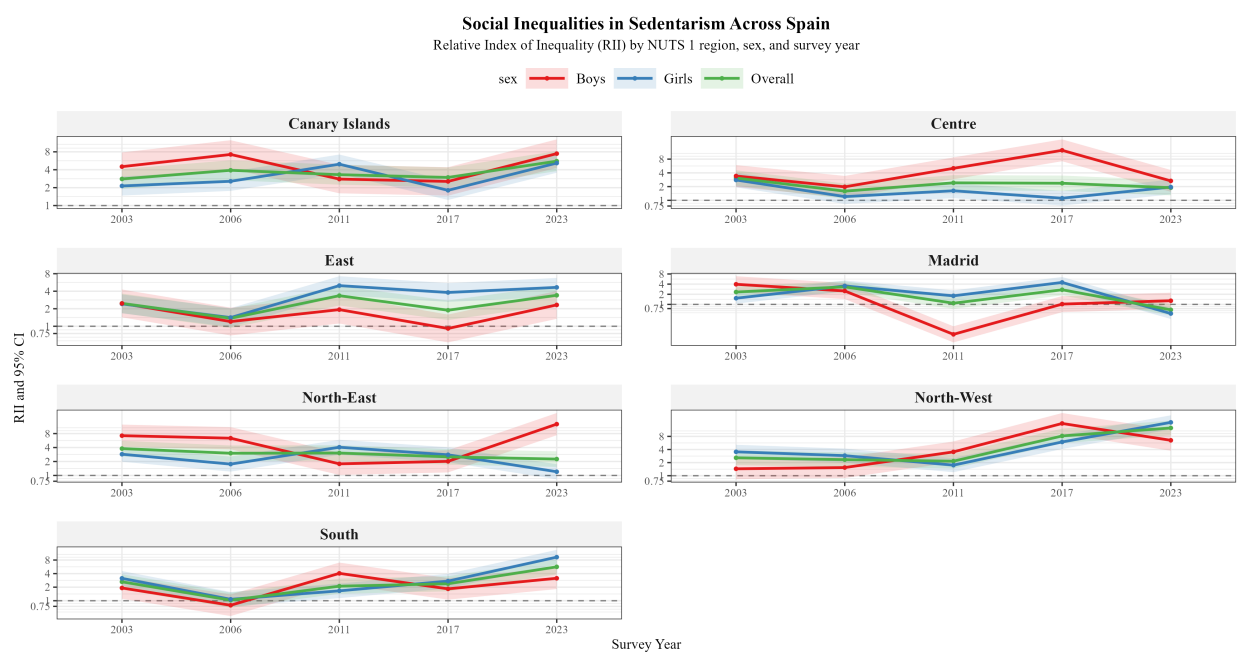


Figure 3: Overall: RII by NUTS per Survey Year

Inequalities in sedentarism across survey years and NUTS regions: In 2003, inequalities were present across all regions, with the highest observed in the North-East at 3.79 (95% CI: 2.58, 5.58), indicating substantially higher prevalence among children in the lowest social class compared to the highest. In contrast, Madrid experienced a marked decline in inequality over time, with the RII decreasing from 2.33 (95% CI: 1.58, 3.43) in 2003 to 0.69 (95% CI: 0.47, 1.02) in 2023, suggesting a reversal of the socioeconomic gradient. The North-West showed the greatest inequality overall, reaching an RII of 12.48 (95% CI: 8.49, 18.36) in 2023. Inequalities also increased over time in the Canary Islands at 5.58 (95% CI: 3.79, 8.20) and South regions at 5.65 (95% CI: 3.84, 8.31) percentage points.

Inequalities in Childhood Sedentarism by NUTS per Survey Year

Unit: Relative Index of Inequality

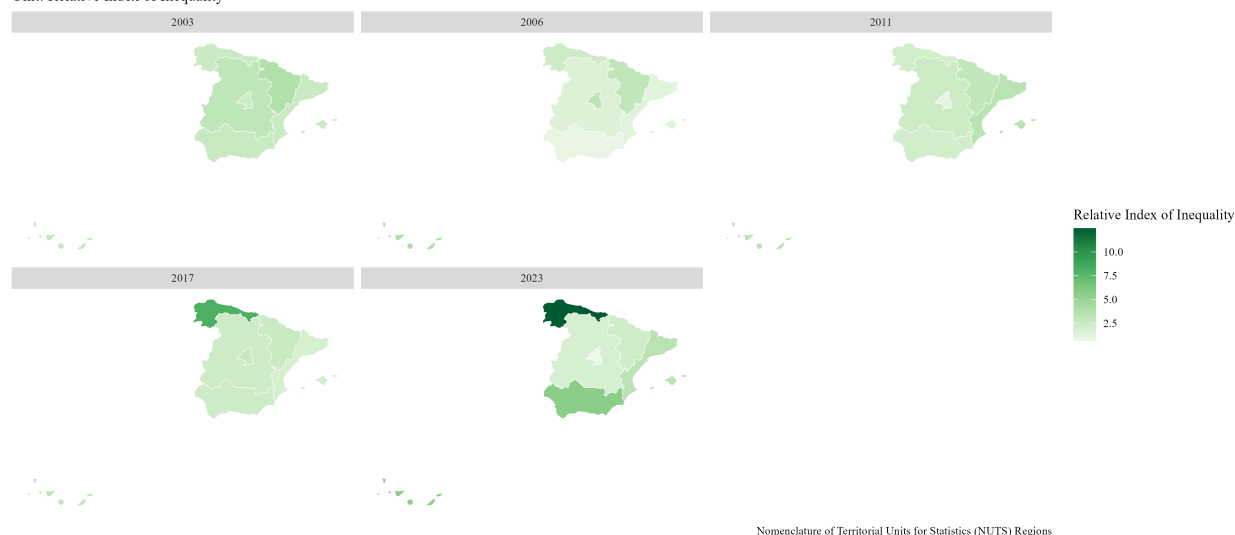


Figure 4: Overall: RII by NUTS per Survey Year

Among girls, relative inequalities in sedentarism were present across all NUTS regions in 2003 and increased in several regions over time. The largest increase was observed in the North-West, where the RII reached 16.79 (95% CI: 11.58, 24.35) in 2023, followed by the South at 9.28 (95% CI: 6.40, 13.45), indicating substantially higher prevalence among girls in the lowest social class compared to the highest.

Among boys, inequalities varied more markedly across regions and survey years. In 2003, the highest inequality was observed in the North-East at 7.21 (95% CI: 4.16, 12.50). Inequalities increased sharply in the Centre and North-West in 2017 with 12.48 (95% CI: 7.20, 21.64) and 15.85 (95% CI: 9.14, 27.48) percentage points, respectively, before declining in 2023. In contrast, the North-East showed a substantial increase by 2023, reaching 12.80 (95% CI: 7.38, 22.19).

ICCs could not be reliably estimated for models including random slopes for social class due to singularity issues. Using random-intercept models only, ICC values ranged from approximately 1% to 5%, indicating minimal regional clustering in sedentarism.

Annual Percent Change and Inflection Point: National: From 2003 to 2023, the annual percent change in relative socioeconomic inequalities of sedentarism at the national level was 5.36% (95% CI: 2.57, 8.22). Among girls, the increase was slightly higher at 5.92% (95% CI: 2.39, 9.56), and among boys slightly lower at 4.64% (95% CI: 1.12, 8.29).

NUTS: Relative inequalities decreased in the North-East (-2.47, 95% CI: -3.12, -1.81) but increased substantially in the North-West (9.33, 95% CI: 4.85, 13.99). Among girls, inequalities increased over time in the

Girls: Inequalities in Childhood Sedentarism by NUTS per Survey Year

Unit: Relative Index of Inequality

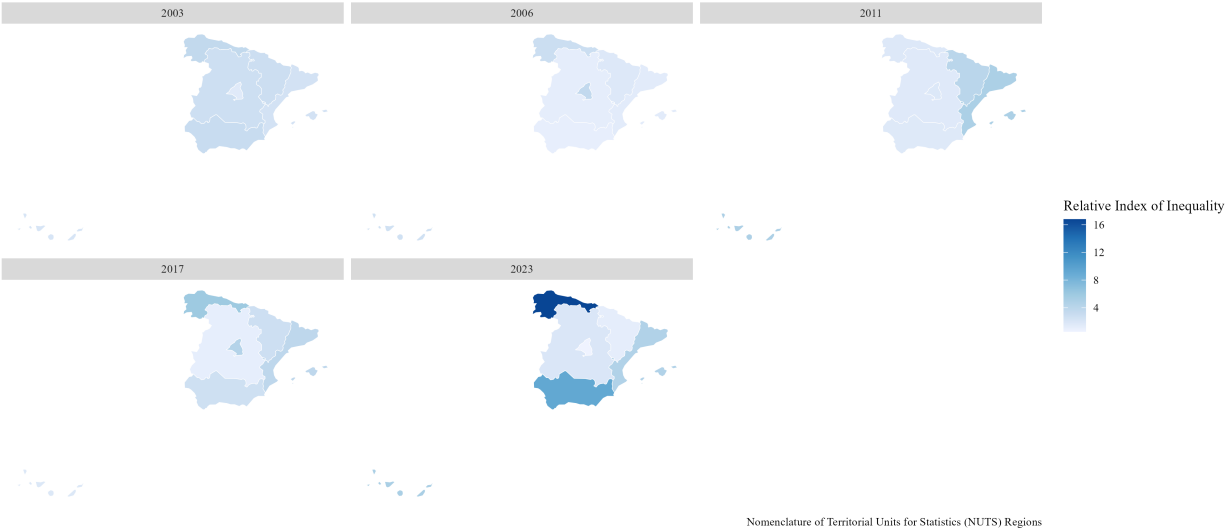


Figure 5: Girls: RII by NUTS per Survey Year

Boys: Inequalities in Childhood Sedentarism by NUTS per Survey Year

Unit: Relative Index of Inequality

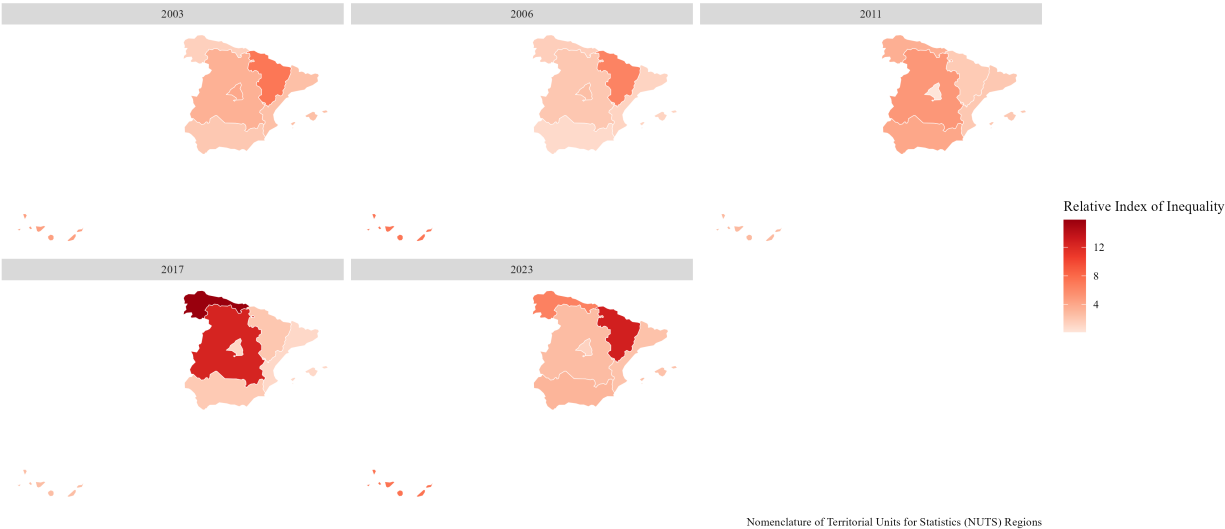


Figure 6: Boys: RII by NUTS per Survey Year

Canary Islands (4.21, 95% CI: 2.33, 6.13) and East regions (3.34, 95% CI: 3.14, 3.54). While among boys, inequalities increased primarily in the North-West (9.57, 95% CI: 2.90, 16.68).

The segmented regression identified an approximate inflection point around 2014, situated between 2011 and 2017 survey years. Prior to this year, the slope of the trend in RII was -0.009 per year, indicating a relatively stable or slightly decreasing inequalities. After this point, the slope increased sharply (0.36 per year).